Data and Artificial Intelligence

Cyber Shujaa Program

Week 4 Assignment
Hotel Revenue Analysis Dashboard Report (Power BI)

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1 Introduction

The objective of this project was to analyze hotel performance data using Microsoft Power BI and create an interactive dashboard to support business decision-making. The analysis focuses

on key hospitality metrics such as Revenue, RevPAR (Revenue per Available Room), ADR (Average Daily Rate), Occupancy Rate, and Realization Rate.

The dashboard enables hotel management to monitor performance across cities, room classes, and time periods, allowing them to identify trends, evaluate property performance, and optimize **revenue strategies.**

2Understanding the Hotel Business Metrics

The hotel industry relies on several performance indicators:

Revenue
Total income generated from room bookings.

RevPAR (Revenue per Available Room)
Measures how effectively rooms generate revenue.

ADR (Average Daily Rate)
Average revenue earned per occupied room.

Occupancy Rate (%)
Percentage of available rooms that were occupied.

Realization (%)

Indicates the percentage of successful bookings compared to total booking attempts.

DSRN (Daily Sellable Room Nights)
Number of rooms available for sale per day.

DBRN (Daily Booked Room Nights)
Number of rooms booked per day.

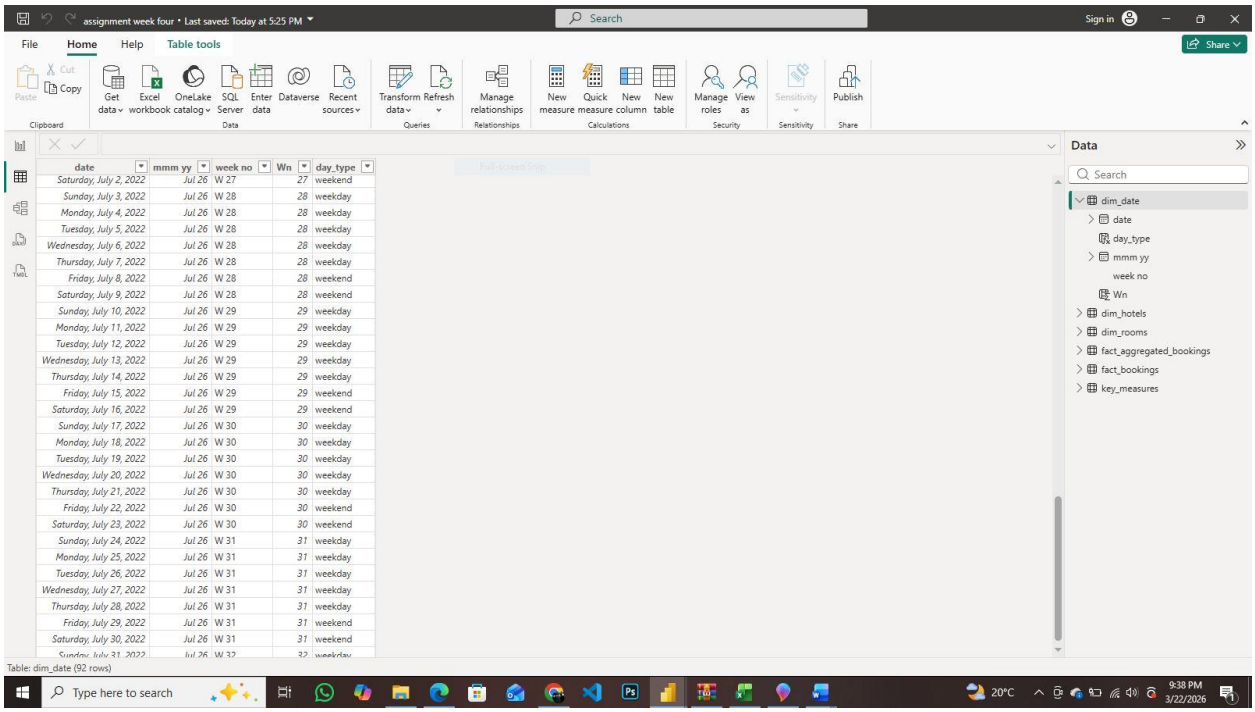
DURN (Daily Utilized Room Nights)
Number of rooms actually occupied.

Tasks Completed

3.data loading and transforation

Data was cleaned and prepared using **Power Query Editor**:

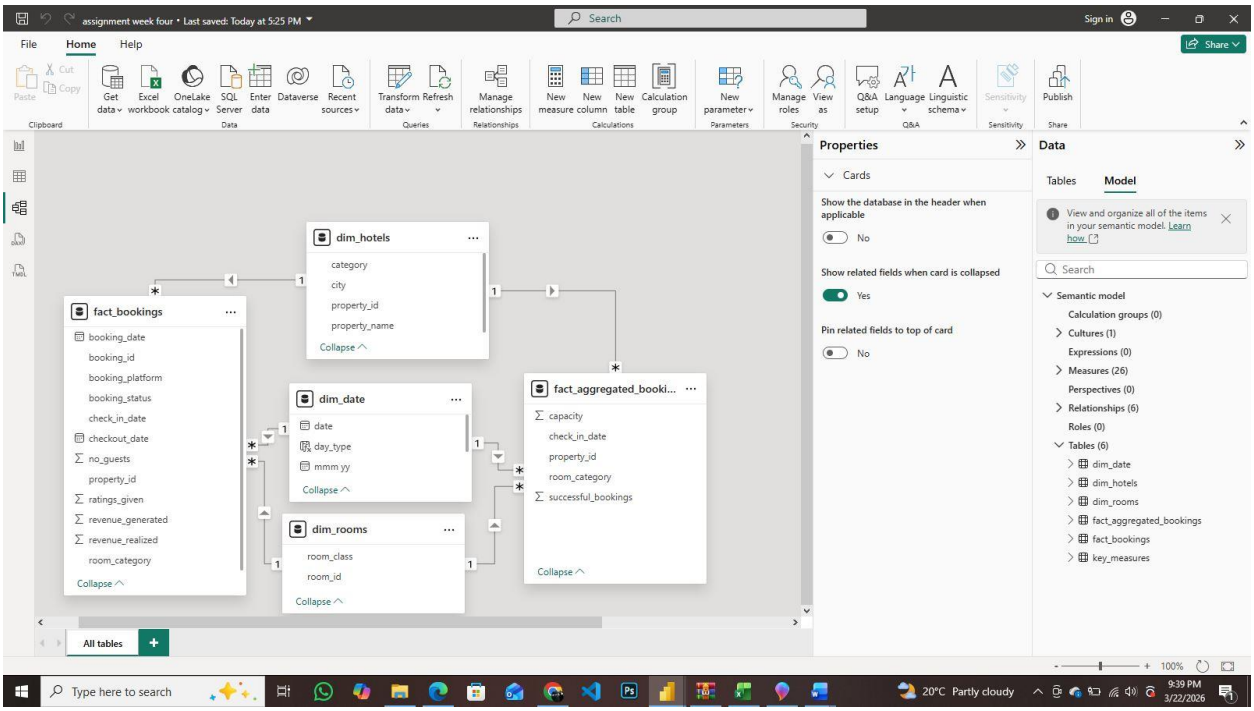
- Removed duplicate records
- Corrected data types
- Handled missing values
- Created calculated columns
- Standardized date formats



date	mmm yy	week no	Wn	day_type
Saturday, July 2, 2022	Jul 26	W 27	27	weekend
Sunday, July 3, 2022	Jul 26	W 28	28	weekday
Monday, July 4, 2022	Jul 26	W 28	28	weekday
Tuesday, July 5, 2022	Jul 26	W 28	28	weekday
Wednesday, July 6, 2022	Jul 26	W 28	28	weekday
Thursday, July 7, 2022	Jul 26	W 28	28	weekday
Friday, July 8, 2022	Jul 26	W 28	28	weekend
Saturday, July 9, 2022	Jul 26	W 28	28	weekend
Sunday, July 10, 2022	Jul 26	W 29	29	weekday
Monday, July 11, 2022	Jul 26	W 29	29	weekday
Tuesday, July 12, 2022	Jul 26	W 29	29	weekday
Wednesday, July 13, 2022	Jul 26	W 29	29	weekday
Thursday, July 14, 2022	Jul 26	W 29	29	weekday
Friday, July 15, 2022	Jul 26	W 29	29	weekend
Saturday, July 16, 2022	Jul 26	W 29	29	weekend
Sunday, July 17, 2022	Jul 26	W 30	30	weekday
Monday, July 18, 2022	Jul 26	W 30	30	weekday
Tuesday, July 19, 2022	Jul 26	W 30	30	weekday
Wednesday, July 20, 2022	Jul 26	W 30	30	weekday
Thursday, July 21, 2022	Jul 26	W 30	30	weekday
Friday, July 22, 2022	Jul 26	W 30	30	weekend
Saturday, July 23, 2022	Jul 26	W 30	30	weekend
Sunday, July 24, 2022	Jul 26	W 31	31	weekday
Monday, July 25, 2022	Jul 26	W 31	31	weekday
Tuesday, July 26, 2022	Jul 26	W 31	31	weekday
Wednesday, July 27, 2022	Jul 26	W 31	31	weekday
Thursday, July 28, 2022	Jul 26	W 31	31	weekday
Friday, July 29, 2022	Jul 26	W 31	31	weekend
Saturday, July 30, 2022	Jul 26	W 31	31	weekend
Sunday, July 31, 2022	Jul 26	W 32	32	weekday

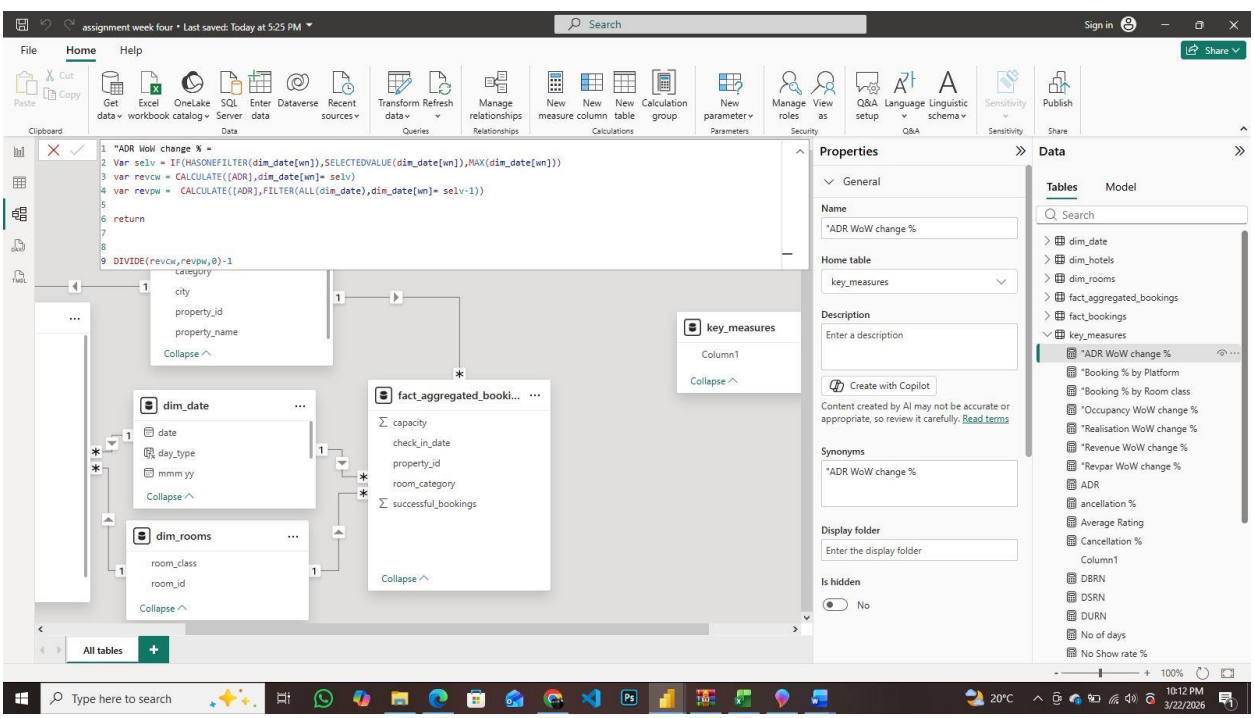
4.data modelling and star achema

- A **Star Schema** was used to structure the data model.



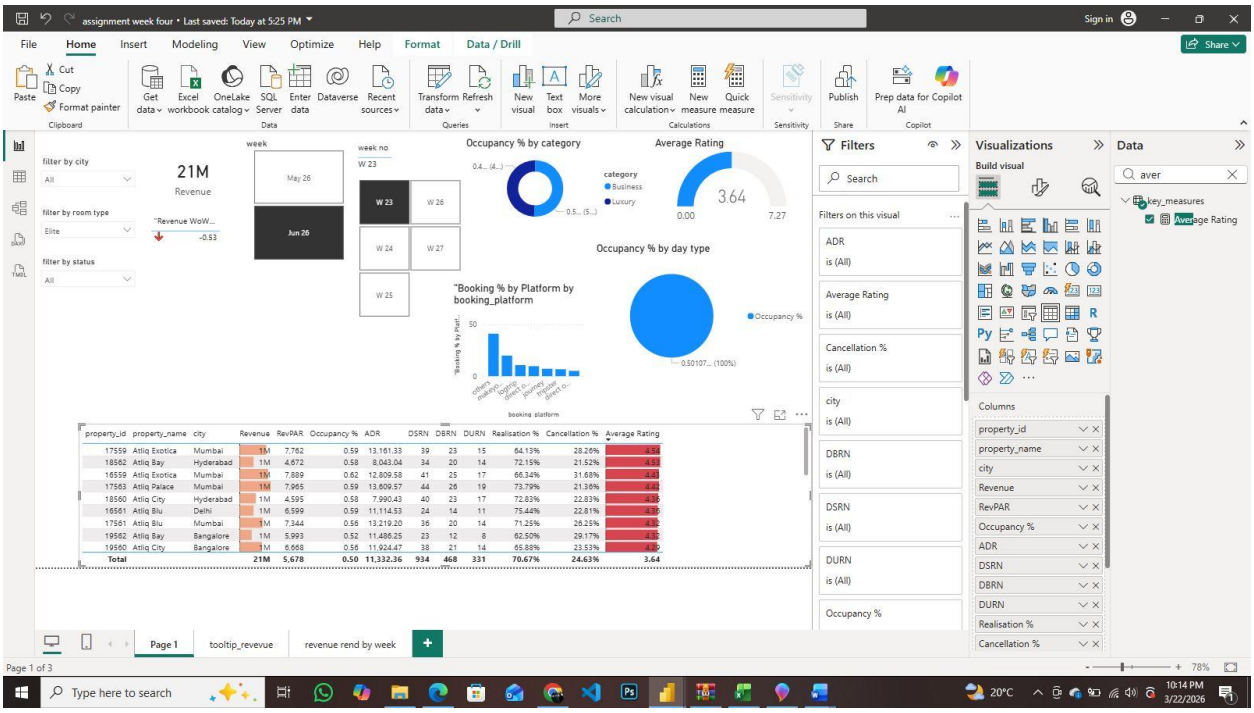
5.data anlysis expresion

- Several **DAX measures and calculated columns** were created to support analysis.



6.dashboard visualization

The Power BI dashboard provides a comprehensive overview of hotel performance



7.publishing and sharing my work

Link to

Code: https://drive.google.com/drive/folders/1zBD7qSRX2w0aUIvsgAfhlEFPTdXsRRlD?usp=sharing

8. Conclusion

This week I gained a good grounding on the introductory concepts relating to data science and artificial intelligence. I am getting a better understanding that I can build on as we work on more advanced concepts in later weeks. I have posted my writeup on my blog and I look forward to building a portfolio that I can showcase on my CV as I look for jobs in Data and AI.